

RAWSON LAKE, ON, Canada

WMO: 718550

Lat: 49.660N

Lon: 93.728W

Elev: 435

StdP: 96.21

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.7	-28.2	-34.3	0.2	-30.5	-31.6	0.2	-28.1	5.9	-11.4	5.4	-11.0	1.6	230	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	28.7	19.5	27.0	18.6	25.4	17.7	21.6	25.8	20.5	24.6	19.4	23.4	3.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.2	15.6	23.8	19.0	14.6	22.9	17.9	13.6	21.5	65.2	25.9	61.2	24.4	57.3	23.5	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.3	4.8	DB	-33.3	31.5	2.1	1.3	-34.9	32.4	-36.1	33.2	-37.3	33.9	-38.9	34.9	(4)
(5)			WB	-33.6	23.2	2.1	0.9	-35.1	23.9	-36.3	24.4	-37.5	24.9	-39.0	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-14.4	-14.4	-4.3	3.3	10.7	16.4	19.6	18.2	12.7	5.2	-3.1	-12.0		
	DBStd	13.38	7.87	6.79	7.22	6.19	5.02	3.30	3.05	3.11	4.65	4.89	6.15	7.12		
	HDD10.0	3413	758	683	444	210	53	1	0	0	23	165	393	682		
	HDD18.3	5629	1016	917	701	451	239	75	22	41	178	407	643	940		
	CDD10.0	955	0	0	2	9	76	193	298	256	104	17	1	0		
	CDD18.3	128	0	0	0	0	4	17	62	37	8	0	0	0		
	CDH23.3	901	0	0	0	0	39	128	432	239	61	1	0	0		
	CDH26.7	173	0	0	0	0	3	17	101	44	8	0	0	0		
(14) Wind	WSAvg	2.4	2.3	2.3	2.5	2.6	2.6	2.3	2.3	2.2	2.4	2.5	2.5	2.2		
(15) Precipitation	PrecAvg	683	31	23	41	41	65	104	87	86	87	57	49	29		
	PrecMax	871	68	37	80	112	144	172	180	167	164	147	158	56		
	PrecMin	508	6	4	6	0	20	46	20	27	14	19	16	20		
	PrecStd	117	15	7	19	34	36	42	45	36	42	36	31	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.8	2.2	17.6	20.6	27.0	28.8	31.5	30.1	27.8	21.7	13.1	3.5		
		MCWB	1.2	-0.5	12.3	10.6	16.1	19.3	20.9	20.6	18.3	14.5	8.6	1.5		
	2%	DB	0.4	0.0	11.4	17.5	24.3	26.4	29.3	27.9	25.1	18.5	8.9	0.1		
		MCWB	-0.1	-1.5	7.1	8.8	14.4	17.2	19.9	19.7	17.7	12.9	5.7	-0.6		
	5%	DB	-1.3	-2.1	7.9	15.0	22.0	24.7	27.5	25.8	22.4	15.2	6.7	-0.9		
		MCWB	-2.2	-3.2	4.2	7.6	13.2	16.4	19.2	18.7	16.4	10.7	4.7	-1.7		
	10%	DB	-3.3	-4.4	4.9	12.6	19.3	23.0	25.7	24.3	20.1	12.7	4.8	-2.3		
		MCWB	-4.0	-5.4	1.9	6.5	12.2	16.1	18.3	18.0	15.1	9.4	3.3	-2.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	0.7	13.5	11.7	18.2	22.3	23.3	22.3	20.8	15.1	9.0	1.3		
		MCDB	2.5	1.8	15.2	17.7	23.5	25.9	28.5	27.1	24.8	20.3	12.2	3.2		
	2%	WB	0.0	-1.3	7.1	9.8	16.5	20.0	22.0	21.1	19.2	13.2	6.6	-0.2		
		MCDB	0.5	-0.4	11.2	14.8	21.1	23.5	26.0	25.5	22.7	17.9	8.2	0.2		
	5%	WB	-2.0	-3.4	4.3	8.4	14.7	18.0	20.7	20.0	17.4	11.4	4.9	-1.6		
		MCDB	-1.3	-2.1	7.3	13.9	19.1	22.1	24.9	24.3	21.4	14.3	6.4	-1.0		
	10%	WB	-3.9	-5.4	2.1	6.9	13.2	16.8	19.7	18.9	15.7	9.6	3.1	-3.0		
		MCDB	-3.2	-4.3	4.4	12.1	18.0	21.1	23.7	23.0	19.3	12.3	4.9	-2.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	10.1	10.4	10.7	10.4	9.6	9.9	9.6	8.6	6.7	6.1	6.8		
		MCDBR	8.3	9.5	11.2	13.6	13.9	12.4	12.0	11.8	11.8	11.0	8.2	6.4		
		MCWBR	7.5	7.8	7.1	7.8	6.9	6.1	5.2	5.5	6.1	6.6	5.8	5.5		
		MCDBR	7.6	8.7	10.2	12.1	11.1	10.0	10.3	10.2	10.3	9.6	7.7	6.0		
	5% WB	MCWBR	7.1	7.3	6.8	7.5	6.8	6.7	5.8	5.8	6.0	6.3	5.6	5.4		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.261	0.286	0.330	0.345	0.365	0.380	0.363	0.343	0.298	0.275	0.248		
	taud		2.346	2.332	2.394	2.430	2.457	2.427	2.394	2.476	2.508	2.608	2.523	2.401		
	Ebn at Noon		812	898	931	918	914	894	872	874	858	854	792	774		
	Edh at Noon		65	84	96	104	106	110	112	99	87	66	56	53		
(44) All-Sky Solar Radiation	RadAvg		1.20	2.40	3.66	4.88	5.34	5.55	5.78	4.84	3.51	2.02	1.08	0.86		
	RadStd		0.14	0.21	0.30	0.38	0.36	0.45	0.39	0.40	0.32	0.27	0.13	0.11		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page